

VZ class project CRW KDE script

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)

suppressWarnings(suppressPackageStartupMessages({
  library(tidyverse)
  library(readxl)
  library(stringr)
  library(lubridate)
  library(purrr)
  library(brms)
  library(cmdstanr)
  library(posterior)
  library(furrr)
  library(future)
  library(maptiles)
}))

project_dir <- "C:/Users/vezuc/OneDrive - Florida International University/PhD/Classes/Bayes
```

Read data

```
gps_2s_clean <- read.csv(
  file.path(project_dir, "gps_2s_clean.csv"),
  stringsAsFactors = FALSE
)
```

Stan model with one track only

```

##prepare
library(dplyr)
library(sf)
library(cmdstanr)
library(posterior)
library(ggplot2)

one_file <- gps_2s_clean %>%
  filter(file_name == unique(gps_2s_clean$file_name)[1]) %>%
  arrange(elapsed_sec) %>%
  slice(1:300)

## convert to sf and project to meters
one_file_sf <- one_file %>%
  st_as_sf(coords = c("longitude", "latitude"), crs = 4326, remove = FALSE) %>%
  st_transform(32725)

## Extract x/y in meters
coords_m <- st_coordinates(one_file_sf)

one_file <- one_file %>%
  mutate(
    x_m = coords_m[, 1],
    y_m = coords_m[, 2]
  )

## center the meter coordinates
one_file <- one_file %>%
  mutate(
    x_obs = x_m - mean(x_m),
    y_obs = y_m - mean(y_m)
  )

## build Stan data
track_dat <- list(
  N = nrow(one_file),
  x_obs = one_file$x_obs,
  y_obs = one_file$y_obs
)

## CRW Stan model in meters
writeLines("

```

```

data {
  int<lower=3> N;
  vector[N] x_obs;
  vector[N] y_obs;
}
parameters {
  vector[N] x_true;
  vector[N] y_true;
  real<lower=0, upper=1> phi;
  real<lower=0> sigma_gps;
  real<lower=0> sigma_move;
}
transformed parameters {
  vector[N-1] dx_true;
  vector[N-1] dy_true;

  for (t in 2:N) {
    dx_true[t-1] = x_true[t] - x_true[t-1];
    dy_true[t-1] = y_true[t] - y_true[t-1];
  }
}
model {
  phi ~ beta(2, 2);

  sigma_gps ~ exponential(1);
  sigma_move ~ exponential(1);

  x_true[1] ~ normal(x_obs[1], 1);
  y_true[1] ~ normal(y_obs[1], 1);

  x_true[2] ~ normal(x_obs[2], 1);
  y_true[2] ~ normal(y_obs[2], 1);

  for (t in 2:(N-1)) {
    dx_true[t] ~ normal(phi * dx_true[t-1], sigma_move);
    dy_true[t] ~ normal(phi * dy_true[t-1], sigma_move);
  }

  x_obs ~ normal(x_true, sigma_gps);
  y_obs ~ normal(y_true, sigma_gps);
}
generated quantities {

```

```

vector[N] x_rep;
vector[N] y_rep;

for (t in 1:N) {
  x_rep[t] = normal_rng(x_true[t], sigma_gps);
  y_rep[t] = normal_rng(y_true[t], sigma_gps);
}
}
", "shark_crw_meters.stan")

## Fit the model
mod_crw <- cmdstan_model("shark_crw_meters.stan")

fit_crw <- mod_crw$sample(
  data = track_dat,
  chains = 4,
  iter_warmup = 1000,
  iter_sampling = 1000,
  refresh = 0
)

```

Running MCMC with 4 sequential chains...

Chain 1 finished in 1.5 seconds.

Chain 2 finished in 1.7 seconds.

Chain 3 finished in 1.8 seconds.

Chain 4 finished in 1.4 seconds.

All 4 chains finished successfully.

Mean chain execution time: 1.6 seconds.

Total execution time: 7.9 seconds.

Batch processing using meters-based CRW Stan model

```

library(dplyr)
library(sf)
library(posterior)

```

```

library(purrr)
library(readr)
library(tidyr)

fit_one_crw_track <- function(df_track, mod_crw) {

  df_track <- df_track %>%
    arrange(elapsed_sec)

  if (nrow(df_track) < 3) return(NULL)

  track_sf <- df_track %>%
    st_as_sf(coords = c("longitude", "latitude"), crs = 4326, remove = FALSE) %>%
    st_transform(32725)

  coords_m <- st_coordinates(track_sf)

  df_track <- df_track %>%
    mutate(
      x_m = coords_m[, 1],
      y_m = coords_m[, 2],
      x_center = x_m - mean(x_m),
      y_center = y_m - mean(y_m)
    )

  track_dat <- list(
    N = nrow(df_track),
    x_obs = df_track$x_center,
    y_obs = df_track$y_center
  )

  fit <- mod_crw$sample(
    data = track_dat,
    chains = 4,
    iter_warmup = 1000,
    iter_sampling = 1000,
    refresh = 0
  )

  x_post <- fit$draws("x_true", format = "matrix")
  y_post <- fit$draws("y_true", format = "matrix")

```

```

x_mean <- colMeans(x_post) + mean(df_track$x_m)
y_mean <- colMeans(y_post) + mean(df_track$y_m)

corrected_sf <- st_as_sf(
  tibble(x_true_m = x_mean, y_true_m = y_mean),
  coords = c("x_true_m", "y_true_m"),
  crs = 32725,
  remove = FALSE
) %>%
  st_transform(4326)

coords_ll <- st_coordinates(corrected_sf)

param_sum <- fit$summary(variables = c("phi", "sigma_gps", "sigma_move")) %>%
  select(variable, mean) %>%
  pivot_wider(names_from = variable, values_from = mean)

out <- df_track %>%
  mutate(
    x_true_m = x_mean,
    y_true_m = y_mean,
    longitude_true = coords_ll[, 1],
    latitude_true = coords_ll[, 2],
    correction_dist_m = sqrt((x_m - x_true_m)^2 + (y_m - y_true_m)^2),
    phi = param_sum$phi,
    sigma_gps = param_sum$sigma_gps,
    sigma_move = param_sum$sigma_move
  )

out
}

dir.create("crw_outputs", showWarnings = FALSE)
dir.create("crw_outputs/track_files", showWarnings = FALSE)
dir.create("crw_outputs/batch_files", showWarnings = FALSE)

track_list <- gps_2s_clean %>%
  group_by(file_name) %>%
  group_split()

batch_size <- 10
track_indices <- seq_along(track_list)

```

```

batch_ids <- split(track_indices, ceiling(seq_along(track_indices) / batch_size))

all_batch_files <- character()

for (b in seq_along(batch_ids)) {

  cat("Starting batch", b, "of", length(batch_ids), "\n")

  this_batch_idx <- batch_ids[[b]]
  this_batch_tracks <- track_list[this_batch_idx]
  batch_results <- vector("list", length(this_batch_tracks))

  for (i in seq_along(this_batch_tracks)) {

    this_track <- this_batch_tracks[[i]]
    this_name <- this_track$file_name[1]
    safe_name <- make.names(this_name)
    track_outfile <- file.path("crw_outputs/track_files", paste0(safe_name, "_crw.csv"))

    if (file.exists(track_outfile)) {
      cat(" Skipping already saved track:", this_name, "\n")
      batch_results[[i]] <- read_csv(track_outfile, show_col_types = FALSE)
      next
    }

    cat(" Fitting track", i, "in batch", b, ":", this_name, "\n")

    fit_result <- tryCatch(
      fit_one_crw_track(this_track, mod_crw),
      error = function(e) {
        message(" Failed on track: ", this_name)
        message(" Error: ", e$message)
        return(NULL)
      }
    )

    batch_results[[i]] <- fit_result

    if (!is.null(fit_result)) {
      write_csv(fit_result, track_outfile)
    }
  }
}

```

```

batch_df <- bind_rows(batch_results)

batch_file <- file.path(
  "crw_outputs/batch_files",
  paste0("crw_batch_", sprintf("%03d", b), ".csv")
)

write_csv(batch_df, batch_file)
all_batch_files <- c(all_batch_files, batch_file)

cat("Finished batch", b, "- saved to:", batch_file, "\n")
}

```

Starting batch 1 of 62

```

Skipping already saved track: 20231226_tuba tigre_DJI_0908_FB_FB.SRT
Skipping already saved track: 20231226_tuba tigre_DJI_0909_FB_FB.SRT
Skipping already saved track: 20231226_tuba tigre_DJI_0910_FB_FB.SRT
Skipping already saved track: 20231226_tuba tigre_DJI_0911_FB_FB.SRT
Skipping already saved track: 20231226_tuba tigre_DJI_0912_FB_FB.SRT
Skipping already saved track: 20231226_tuba tigre_DJI_0913_FB_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0966_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0967_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0968_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0969_FB.SRT

```

Finished batch 1 - saved to: crw_outputs/batch_files/crw_batch_001.csv

Starting batch 2 of 62

```

Skipping already saved track: 20240104_tuba tigre_DJI_0971_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0972_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0973_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0974_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0975_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0976_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0977_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0978_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0979_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0980_FB.SRT

```

Finished batch 2 - saved to: crw_outputs/batch_files/crw_batch_002.csv

Starting batch 3 of 62

```

Skipping already saved track: 20240104_tuba tigre_DJI_0981_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0982_FB.SRT
Skipping already saved track: 20240104_tuba tigre_DJI_0983_FB.SRT

```


Skipping already saved track: 20240104_tuba tigre_DJI_0984_FB.SRT
 Skipping already saved track: 20240112_tuba tigre_DJI_0058_FB.SRT
 Skipping already saved track: 20240112_tuba tigre_DJI_0059_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0144_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0145_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0147_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0148_FB.SRT
 Finished batch 3 - saved to: crw_outputs/batch_files/crw_batch_003.csv
 Starting batch 4 of 62
 Skipping already saved track: 20240120_tuba tigre_DJI_0149_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0150_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0151_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0152_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0153_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0154_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0155_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0156_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0157_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0158_FB.SRT
 Finished batch 4 - saved to: crw_outputs/batch_files/crw_batch_004.csv
 Starting batch 5 of 62
 Skipping already saved track: 20240120_tuba tigre_DJI_0159_FB.SRT
 Skipping already saved track: 20240120_tuba tigre_DJI_0160_FB.SRT
 Skipping already saved track: 20240126_tuba tigre_DJI_0208_FB.SRT
 Skipping already saved track: 20240126_tuba tigre_DJI_0209_FB.SRT
 Skipping already saved track: 20240126_tuba tigre_DJI_0210_FB.SRT
 Skipping already saved track: 20240207_tuba tigre_DJI_0274_FB.SRT
 Skipping already saved track: 20240207_tuba tigre_DJI_0275_FB.SRT
 Skipping already saved track: 20240207_tuba tigre_DJI_0276_FB.SRT
 Skipping already saved track: 20240207_tuba tigre_DJI_0277_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0594_FB.SRT
 Finished batch 5 - saved to: crw_outputs/batch_files/crw_batch_005.csv
 Starting batch 6 of 62
 Skipping already saved track: 20240628_tuba tigre_DJI_0595_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0596_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0597_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0598_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0599_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0603_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0604_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0605_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0606_FB.SRT
 Skipping already saved track: 20240628_tuba tigre_DJI_0607_FB.SRT

Finished batch 6 - saved to: crw_outputs/batch_files/crw_batch_006.csv
Starting batch 7 of 62
Skipping already saved track: 20240713_tuba tigre_DJI_0653_FB.SRT
Skipping already saved track: 20240713_tuba tigre_DJI_0655_FB.SRT
Skipping already saved track: 20240713_tuba tigre_DJI_0656_FB.SRT
Skipping already saved track: 20240713_tuba tigre_DJI_0657_FB.SRT
Skipping already saved track: 20240713_tuba tigre_DJI_0658_FB.SRT
Skipping already saved track: 20240713_tuba tigre_tuba limao_DJI_0652_FB.SRT
Skipping already saved track: 20240731_tuba tigre_DJI_0758_FB.SRT
Skipping already saved track: 20240731_tuba tigre_DJI_0760_FB.SRT
Skipping already saved track: 20240731_tuba tigre_DJI_0761_FB.SRT
Skipping already saved track: 20240731_tuba tigre_DJI_0762_FB.SRT
Finished batch 7 - saved to: crw_outputs/batch_files/crw_batch_007.csv
Starting batch 8 of 62
Skipping already saved track: 20240731_tuba tigre_DJI_0763_FB.SRT
Skipping already saved track: 20240820_tuba tigre_DJI_0871_FB.SRT
Skipping already saved track: 20240820_tuba tigre_DJI_0872_FB.SRT
Skipping already saved track: 20240829_tuba tigre_DJI_0912_FB.SRT
Skipping already saved track: 20240829_tuba tigre_DJI_0913_FB.SRT
Skipping already saved track: 20240829_tuba tigre_DJI_0914_FB.SRT
Skipping already saved track: 20240829_tuba tigre_DJI_0915_FB.SRT
Skipping already saved track: 20240829_tuba tigre_DJI_0916_FB.SRT
Skipping already saved track: 20240829_tuba tigre_DJI_0917_FB.SRT
Skipping already saved track: 20240829_tuba tigre_DJI_0918_FB.SRT
Finished batch 8 - saved to: crw_outputs/batch_files/crw_batch_008.csv
Starting batch 9 of 62
Fitting track 1 in batch 9 : 20240829_tuba tigre_DJI_0919_FB.SRT
Skipping already saved track: 20240831_tuba tigre_DJI_0937_FB.SRT
Skipping already saved track: 20240831_tuba tigre_DJI_0938_FB.SRT
Skipping already saved track: 20240831_tuba tigre_DJI_0939_FB.SRT
Skipping already saved track: 20240831_tuba tigre_DJI_0940_FB.SRT
Skipping already saved track: 20240831_tuba tigre_DJI_0941_FB.SRT
Skipping already saved track: 20240831_tuba tigre_DJI_0942_FB.SRT
Skipping already saved track: 20240831_tuba tigre_DJI_0943_FB.SRT
Skipping already saved track: 20240831_tuba tigre_DJI_0944_FB.SRT
Skipping already saved track: 20240831_tuba tigre_DJI_0945_FB.SRT
Finished batch 9 - saved to: crw_outputs/batch_files/crw_batch_009.csv
Starting batch 10 of 62
Skipping already saved track: 20240831_tuba tigre_DJI_0949_FB.SRT
Skipping already saved track: 20240905_tuba tigre_DJI_0961_FB.SRT
Skipping already saved track: 20240905_tuba tigre_DJI_0962_FB.SRT
Skipping already saved track: 20240905_tuba tigre_DJI_0963_FB.SRT
Skipping already saved track: 20240905_tuba tigre_DJI_0964_FB.SRT

Skipping already saved track: 20240907_tuba tigre mercado_DJI_0976_FB.SRT
 Skipping already saved track: 20240907_tuba tigre mercado_DJI_0977_FB.SRT
 Skipping already saved track: 20240907_tuba tigre mercado_DJI_0978_FB.SRT
 Skipping already saved track: 20240907_tuba tigre mercado_DJI_0979_FB.SRT
 Skipping already saved track: 20240907_tuba tigre mercado_DJI_0980_FB.SRT
 Finished batch 10 - saved to: crw_outputs/batch_files/crw_batch_010.csv
 Starting batch 11 of 62
 Skipping already saved track: 20240907_tuba tigre mercado_DJI_0981_FB.SRT
 Skipping already saved track: 20240907_tuba tigre mercado_DJI_0982_FB.SRT
 Skipping already saved track: 20240907_tuba tigre mercado_DJI_0983_FB.SRT
 Skipping already saved track: 20240928_tuba tigre_DJI_0097_FB.SRT
 Skipping already saved track: 20240928_tuba tigre_DJI_0098_FB.SRT
 Skipping already saved track: 20240928_tuba tigre_DJI_0099_FB.SRT
 Skipping already saved track: 20240928_tuba tigre_DJI_0100_FB.SRT
 Skipping already saved track: 20240928_tuba tigre_DJI_0101_FB.SRT
 Skipping already saved track: 202412_tuba tigre_DJI_0956_FB.SRT
 Skipping already saved track: 202412_tuba tigre_DJI_0957_FB.SRT
 Finished batch 11 - saved to: crw_outputs/batch_files/crw_batch_011.csv
 Starting batch 12 of 62
 Skipping already saved track: 202412_tuba tigre_DJI_0958_FB.SRT
 Skipping already saved track: 202412_tuba tigre_DJI_0959_FB.SRT
 Skipping already saved track: 202412_tuba tigre_DJI_0961_FB.SRT
 Skipping already saved track: CAM_20260116111515_0007_V.SRT
 Skipping already saved track: CAM_20260116111541_0008_V.SRT
 Skipping already saved track: CAM_20260116111620_0009_V.SRT
 Skipping already saved track: CAM_20260116111656_0010_V.SRT
 Skipping already saved track: CAM_20260116111806_0011_V.SRT
 Skipping already saved track: CAM_20260116111922_0012_V.SRT
 Skipping already saved track: CAM_20260116153342_0008_V.SRT
 Finished batch 12 - saved to: crw_outputs/batch_files/crw_batch_012.csv
 Starting batch 13 of 62
 Skipping already saved track: CAM_20260116153429_0009_V.SRT
 Skipping already saved track: CAM_20260116153532_0010_V.SRT
 Skipping already saved track: CAM_20260116153614_0011_V.SRT
 Skipping already saved track: CAM_20260116153712_0012_V.SRT
 Skipping already saved track: CAM_20260117111401_0002_V.SRT
 Skipping already saved track: CAM_20260117111426_0003_V.SRT
 Skipping already saved track: CAM_20260117111457_0004_V.SRT
 Skipping already saved track: CAM_20260117111846_0005_V.SRT
 Skipping already saved track: CAM_20260117112114_0006_V.SRT
 Skipping already saved track: CAM_20260120153904_0001_V.SRT
 Finished batch 13 - saved to: crw_outputs/batch_files/crw_batch_013.csv
 Starting batch 14 of 62

Skipping already saved track: CAM_20260120161022_0001_V.SRT
 Skipping already saved track: CAM_20260125091424_0001_V.SRT
 Skipping already saved track: CAM_20260125091634_0002_V.SRT
 Skipping already saved track: CAM_20260125092023_0003_V.SRT
 Skipping already saved track: CAM_20260125092107_0004_V.SRT
 Skipping already saved track: CAM_20260125092456_0005_V.SRT
 Skipping already saved track: CAM_20260125092526_0006_V.SRT
 Skipping already saved track: CAM_20260125092549_0007_V.SRT
 Skipping already saved track: CAM_20260125092747_0008_V.SRT
 Skipping already saved track: CAM_20260125093136_0009_V.SRT
 Finished batch 14 - saved to: crw_outputs/batch_files/crw_batch_014.csv
 Starting batch 15 of 62
 Skipping already saved track: CAM_20260125093401_0010_V.SRT
 Skipping already saved track: CAM_20260125093442_0011_V.SRT
 Skipping already saved track: CAM_20260206081124_0001_V.SRT
 Skipping already saved track: CAM_20260206081408_0002_V.SRT
 Skipping already saved track: CAM_20260206081756_0003_V.SRT
 Skipping already saved track: CAM_20260206082132_0004_V.SRT
 Skipping already saved track: CAM_20260206082746_0001_V.SRT
 Skipping already saved track: CAM_20260206083134_0002_V.SRT
 Skipping already saved track: CAM_20260206083523_0003_V.SRT
 Skipping already saved track: CAM_20260206083846_0004_V.SRT
 Finished batch 15 - saved to: crw_outputs/batch_files/crw_batch_015.csv
 Starting batch 16 of 62
 Skipping already saved track: CAM_20260206084237_0005_V.SRT
 Skipping already saved track: CAM_20260206084625_0006_V.SRT
 Skipping already saved track: CAM_20260209085235_0007_V.SRT
 Skipping already saved track: CAM_20260209085634_0001_V.SRT
 Skipping already saved track: CAM_20260209090023_0002_V.SRT
 Skipping already saved track: CAM_20260209090412_0003_V.SRT
 Skipping already saved track: CAM_20260209090800_0004_V.SRT
 Skipping already saved track: CAM_20260209091149_0005_V.SRT
 Skipping already saved track: CAM_20260209091741_0006_V.SRT
 Skipping already saved track: CAM_20260313162142_0002_V.SRT
 Finished batch 16 - saved to: crw_outputs/batch_files/crw_batch_016.csv
 Starting batch 17 of 62
 Skipping already saved track: CAM_20260313162544_0003_V.SRT
 Skipping already saved track: CAM_20260313164914_0001_V.SRT
 Skipping already saved track: CAM_20260313165303_0002_V.SRT
 Skipping already saved track: CAM_20260313165653_0003_V.SRT
 Skipping already saved track: CAM_20260313165907_0004_V.SRT
 Skipping already saved track: CAM_20260313170256_0005_V.SRT
 Skipping already saved track: CAM_20260330110313_0005_V.SRT

Skipping already saved track: CAM_20260330111014_0006_V.SRT
 Skipping already saved track: CAM_20260330111553_0007_V.SRT
 Skipping already saved track: DJI_0387.SRT
 Finished batch 17 - saved to: crw_outputs/batch_files/crw_batch_017.csv
 Starting batch 18 of 62
 Skipping already saved track: DJI_0388.SRT
 Skipping already saved track: DJI_0389.SRT
 Skipping already saved track: DJI_0390.SRT
 Skipping already saved track: DJI_0391.SRT
 Skipping already saved track: DJI_0392.SRT
 Skipping already saved track: DJI_0393.SRT
 Skipping already saved track: DJI_0394.SRT
 Skipping already saved track: DJI_0395.SRT
 Skipping already saved track: DJI_0396.SRT
 Skipping already saved track: DJI_0397.SRT
 Finished batch 18 - saved to: crw_outputs/batch_files/crw_batch_018.csv
 Starting batch 19 of 62
 Skipping already saved track: DJI_0398.SRT
 Skipping already saved track: DJI_0399.SRT
 Skipping already saved track: DJI_20251117110646_0003_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117111610_0004_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117113016_0005_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117113214_0006_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117113334_0007_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117113615_0008_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117113721_0009_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117113822_0010_D_MALIBU4.SRT
 Finished batch 19 - saved to: crw_outputs/batch_files/crw_batch_019.csv
 Starting batch 20 of 62
 Skipping already saved track: DJI_20251117113837_0011_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117113932_0012_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117114124_0013_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251117114223_0014_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251119030819_0006_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251119030904_0007_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251119031004_0008_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251119031634_0009_D_MALIBU4.SRT
 Fitting track 9 in batch 20 : DJI_20251119031852_0010_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251119032847_0011_D_MALIBU4.SRT
 Finished batch 20 - saved to: crw_outputs/batch_files/crw_batch_020.csv
 Starting batch 21 of 62
 Skipping already saved track: DJI_20251119033029_0012_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251119052830_0042_D_MALIBU4.SRT

Skipping already saved track: DJI_20251119055253_0058_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251119101922_0083_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251119102828_0084_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251120021920_0001_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251120022255_0002_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251120022454_0003_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251120022633_0004_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121033332_0001_D_MALIBU4.SRT
 Finished batch 21 - saved to: crw_outputs/batch_files/crw_batch_021.csv
 Starting batch 22 of 62
 Skipping already saved track: DJI_20251121033352_0002_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121033458_0003_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121033746_0004_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121033824_0005_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121033932_0006_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121034001_0007_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121034019_0008_D_MALIBU4.SRT
 Fitting track 8 in batch 22 : DJI_20251121034051_0009_D_MALIBU4.SRT
 Fitting track 9 in batch 22 : DJI_20251121034054_0010_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121034056_0011_D_MALIBU4.SRT
 Finished batch 22 - saved to: crw_outputs/batch_files/crw_batch_022.csv
 Starting batch 23 of 62
 Skipping already saved track: DJI_20251121060124_0041_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121060518_0042_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121060603_0043_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121060713_0044_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121061403_0045_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121061614_0046_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121062550_0047_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121063106_0048_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121074000_0050_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121074221_0051_D_MALIBU4.SRT
 Finished batch 23 - saved to: crw_outputs/batch_files/crw_batch_023.csv
 Starting batch 24 of 62
 Skipping already saved track: DJI_20251121074306_0052_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121092003_0055_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121092109_0056_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121092133_0057_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121092154_0058_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121092242_0059_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121092334_0060_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121092634_0061_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121092802_0062_D_MALIBU4.SRT

Fitting track 10 in batch 24 : DJI_20251121092843_0063_D_MALIBU4.SRT
 Finished batch 24 - saved to: crw_outputs/batch_files/crw_batch_024.csv
 Starting batch 25 of 62
 Skipping already saved track: DJI_20251121092921_0064_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251121093106_0065_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251122103550_0038_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251122103741_0040_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251123101135_0005_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251123101159_0006_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251123101346_0007_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251123101701_0008_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251123101731_0009_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251123101755_0010_D_MALIBU4.SRT
 Finished batch 25 - saved to: crw_outputs/batch_files/crw_batch_025.csv
 Starting batch 26 of 62
 Fitting track 1 in batch 26 : DJI_20251123101819_0011_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124015154_0001_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124015642_0002_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124015735_0003_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124015821_0004_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124015839_0005_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124015950_0006_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124020153_0008_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124020222_0009_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124020346_0010_D_MALIBU4.SRT
 Finished batch 26 - saved to: crw_outputs/batch_files/crw_batch_026.csv
 Starting batch 27 of 62
 Skipping already saved track: DJI_20251124020414_0011_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124022540_0012_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124022841_0013_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124022953_0014_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023008_0015_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023024_0016_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023111_0017_D_MALIBU4.SRT
 Fitting track 8 in batch 27 : DJI_20251124023155_0018_D_MALIBU4.SRT
 Fitting track 9 in batch 27 : DJI_20251124023157_0019_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023204_0020_D_MALIBU4.SRT
 Finished batch 27 - saved to: crw_outputs/batch_files/crw_batch_027.csv
 Starting batch 28 of 62
 Skipping already saved track: DJI_20251124023243_0021_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023337_0022_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023508_0023_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023533_0024_D_MALIBU4.SRT

Skipping already saved track: DJI_20251124023623_0025_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023728_0026_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023809_0027_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023830_0028_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124023919_0029_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124024004_0030_D_MALIBU4.SRT
 Finished batch 28 - saved to: crw_outputs/batch_files/crw_batch_028.csv
 Starting batch 29 of 62
 Skipping already saved track: DJI_20251124024231_0032_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124024329_0033_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124024344_0034_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124024456_0035_D_MALIBU4.SRT
 Fitting track 5 in batch 29 : DJI_20251124024739_0036_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124025621_0037_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124025644_0038_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124030710_0039_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124031217_0044_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124031227_0045_D_MALIBU4.SRT
 Finished batch 29 - saved to: crw_outputs/batch_files/crw_batch_029.csv
 Starting batch 30 of 62
 Skipping already saved track: DJI_20251124031237_0046_D_MALIBU4.SRT
 Fitting track 2 in batch 30 : DJI_20251124031303_0047_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124031305_0048_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124031351_0049_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124031441_0050_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124031530_0051_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124031606_0052_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124031655_0053_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124031721_0054_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124032211_0055_D_MALIBU4.SRT
 Finished batch 30 - saved to: crw_outputs/batch_files/crw_batch_030.csv
 Starting batch 31 of 62
 Skipping already saved track: DJI_20251124032255_0056_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124032317_0057_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124032516_0058_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124032558_0059_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124032704_0060_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124032747_0061_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124033038_0062_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124033102_0063_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251124033237_0064_D_MALIBU4.SRT
 Fitting track 10 in batch 31 : DJI_20251124033527_0065_D_MALIBU4.SRT
 Finished batch 31 - saved to: crw_outputs/batch_files/crw_batch_031.csv

Starting batch 32 of 62

Skipping already saved track: DJI_20251128075354_0047_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128075421_0048_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128075510_0049_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128075813_0050_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128075851_0051_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128080116_0052_D_MALIBU4.SRT

Fitting track 7 in batch 32 : DJI_20251128080148_0053_D_MALIBU4.SRT

Fitting track 8 in batch 32 : DJI_20251128080153_0054_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128084451_0086_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128084530_0087_D_MALIBU4.SRT

Finished batch 32 - saved to: crw_outputs/batch_files/crw_batch_032.csv

Starting batch 33 of 62

Skipping already saved track: DJI_20251128084707_0088_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128090232_0089_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128090256_0090_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128090331_0091_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128090346_0092_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128090435_0093_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128090520_0094_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128090558_0095_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128090705_0096_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128090843_0097_D_MALIBU4.SRT

Finished batch 33 - saved to: crw_outputs/batch_files/crw_batch_033.csv

Starting batch 34 of 62

Skipping already saved track: DJI_20251128091007_0098_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128091526_0099_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128091549_0100_D_MALIBU4.SRT

Skipping already saved track: DJI_20251128091604_0101_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129020053_0001_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129020145_0002_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129020213_0003_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129020248_0004_D_MALIBU4.SRT

Fitting track 9 in batch 34 : DJI_20251129020255_0005_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129020258_0006_D_MALIBU4.SRT

Finished batch 34 - saved to: crw_outputs/batch_files/crw_batch_034.csv

Starting batch 35 of 62

Skipping already saved track: DJI_20251129020528_0007_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129020633_0008_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129020733_0009_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129020830_0010_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129020900_0011_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129021024_0012_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129021047_0013_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129021149_0014_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129021233_0015_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129021335_0016_D_MALIBU4.SRT
 Finished batch 35 - saved to: crw_outputs/batch_files/crw_batch_035.csv
 Starting batch 36 of 62
 Skipping already saved track: DJI_20251129021401_0017_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129021458_0018_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129021549_0019_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129021630_0020_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129021722_0021_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129021822_0022_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129022555_0023_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129044208_0043_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129044312_0044_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129044341_0045_D_MALIBU4.SRT
 Finished batch 36 - saved to: crw_outputs/batch_files/crw_batch_036.csv
 Starting batch 37 of 62
 Skipping already saved track: DJI_20251129044410_0046_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129044648_0047_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129045113_0049_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129045414_0050_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129045527_0051_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129045643_0052_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129045705_0053_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129045804_0054_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129045831_0055_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129045933_0056_D_MALIBU4.SRT
 Finished batch 37 - saved to: crw_outputs/batch_files/crw_batch_037.csv
 Starting batch 38 of 62
 Skipping already saved track: DJI_20251129050319_0057_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129050333_0058_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129050340_0059_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129050752_0061_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129051130_0062_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129051222_0063_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129051816_0064_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129051946_0065_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129052054_0066_D_MALIBU4.SRT
 Fitting track 10 in batch 38 : DJI_20251129052619_0067_D_MALIBU4.SRT
 Finished batch 38 - saved to: crw_outputs/batch_files/crw_batch_038.csv
 Starting batch 39 of 62
 Skipping already saved track: DJI_20251129052622_0068_D_MALIBU4.SRT

Skipping already saved track: DJI_20251129052628_0069_D_MALIBU4.SRT
 Fitting track 3 in batch 39 : DJI_20251129052640_0070_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129052815_0071_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129053345_0072_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129053421_0073_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129053546_0074_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129053802_0075_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129053828_0076_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129053910_0077_D_MALIBU4.SRT
 Finished batch 39 - saved to: crw_outputs/batch_files/crw_batch_039.csv
 Starting batch 40 of 62
 Skipping already saved track: DJI_20251129054012_0078_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129054150_0079_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129054843_0080_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129065507_0001_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129065628_0002_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129102304_0030_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129102808_0031_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129102901_0032_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129102929_0033_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129103021_0034_D_MALIBU4.SRT
 Finished batch 40 - saved to: crw_outputs/batch_files/crw_batch_040.csv
 Starting batch 41 of 62
 Skipping already saved track: DJI_20251129103052_0035_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129103130_0036_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129103428_0037_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129103506_0038_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129103658_0039_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129103747_0040_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129103848_0041_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129103944_0042_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129104348_0043_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129104500_0044_D_MALIBU4.SRT
 Finished batch 41 - saved to: crw_outputs/batch_files/crw_batch_041.csv
 Starting batch 42 of 62
 Skipping already saved track: DJI_20251129104931_0045_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129111725_0046_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129112515_0047_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251129112652_0048_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130022623_0001_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130022701_0002_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130022743_0003_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130023244_0004_D_MALIBU4.SRT

Skipping already saved track: DJI_20251130023336_0005_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130023446_0006_D_MALIBU4.SRT
 Finished batch 42 - saved to: crw_outputs/batch_files/crw_batch_042.csv
 Starting batch 43 of 62
 Skipping already saved track: DJI_20251130023518_0007_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130023555_0008_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130023627_0009_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130023710_0010_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130023755_0011_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130024154_0012_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130024307_0013_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130024332_0014_D_MALIBU4.SRT
 Fitting track 9 in batch 43 : DJI_20251130024345_0015_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130024349_0016_D_MALIBU4.SRT
 Finished batch 43 - saved to: crw_outputs/batch_files/crw_batch_043.csv
 Starting batch 44 of 62
 Skipping already saved track: DJI_20251130024447_0017_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130025243_0018_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130025342_0019_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130025359_0020_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130025417_0021_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130025508_0022_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130025602_0023_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130034139_0027_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130034223_0028_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130034301_0029_D_MALIBU4.SRT
 Finished batch 44 - saved to: crw_outputs/batch_files/crw_batch_044.csv
 Starting batch 45 of 62
 Skipping already saved track: DJI_20251130034401_0030_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130034826_0031_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130035236_0032_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130050256_0038_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130050440_0039_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130050521_0040_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130051210_0041_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251130051531_0042_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206083116_0001_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206083139_0002_D_MALIBU4.SRT
 Finished batch 45 - saved to: crw_outputs/batch_files/crw_batch_045.csv
 Starting batch 46 of 62
 Skipping already saved track: DJI_20251206083223_0003_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206083453_0004_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206083607_0005_D_MALIBU4.SRT

Fitting track 4 in batch 46 : DJI_20251206083654_0006_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206083657_0007_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206083722_0008_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206083821_0009_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206083856_0010_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206083925_0011_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206084016_0012_D_MALIBU4.SRT
 Finished batch 46 - saved to: crw_outputs/batch_files/crw_batch_046.csv
 Starting batch 47 of 62
 Skipping already saved track: DJI_20251206084034_0013_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206084135_0014_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206084241_0015_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206084450_0016_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206084648_0017_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206084707_0018_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206085020_0019_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251206091134_0032_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251207092205_0012_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251207092216_0013_D_MALIBU4.SRT
 Finished batch 47 - saved to: crw_outputs/batch_files/crw_batch_047.csv
 Starting batch 48 of 62
 Skipping already saved track: DJI_20251207092256_0014_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251207092415_0015_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251207092435_0016_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211095804_0030_D_MALIBU4.SRT
 Fitting track 5 in batch 48 : DJI_20251211095851_0031_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211095856_0032_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211095951_0033_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100037_0034_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100105_0035_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100133_0036_D_MALIBU4.SRT
 Finished batch 48 - saved to: crw_outputs/batch_files/crw_batch_048.csv
 Starting batch 49 of 62
 Skipping already saved track: DJI_20251211100154_0037_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100233_0038_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100428_0039_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100501_0040_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100517_0041_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100525_0042_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100611_0043_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100619_0044_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100647_0045_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100709_0046_D_MALIBU4.SRT

Finished batch 49 - saved to: crw_outputs/batch_files/crw_batch_049.csv
Starting batch 50 of 62
 Skipping already saved track: DJI_20251211100826_0047_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100841_0048_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100917_0049_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211100930_0050_D_MALIBU4.SRT
 Fitting track 5 in batch 50 : DJI_20251211101012_0051_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101014_0052_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101044_0053_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101119_0054_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101222_0055_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101252_0056_D_MALIBU4.SRT
Finished batch 50 - saved to: crw_outputs/batch_files/crw_batch_050.csv
Starting batch 51 of 62
 Skipping already saved track: DJI_20251211101314_0057_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101449_0058_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101524_0059_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101558_0060_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101641_0061_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101711_0062_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101742_0063_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211101931_0064_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211102016_0065_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211102111_0066_D_MALIBU4.SRT
Finished batch 51 - saved to: crw_outputs/batch_files/crw_batch_051.csv
Starting batch 52 of 62
 Skipping already saved track: DJI_20251211102241_0067_D_MALIBU4.SRT
 Skipping already saved track: DJI_20251211102401_0068_D_MALIBU4.SRT
 Skipping already saved track: tuba tigre DJI_0120.SRT
 Skipping already saved track: tuba tigre DJI_0121.SRT
 Skipping already saved track: tuba tigre DJI_0122.SRT
 Skipping already saved track: tuba tigre DJI_0123.SRT
 Skipping already saved track: tuba tigre DJI_0124.SRT
 Skipping already saved track: tuba tigre DJI_0137_101.SRT
 Skipping already saved track: tuba tigre DJI_0138_101.SRT
 Skipping already saved track: tuba tigre DJI_0435.SRT
Finished batch 52 - saved to: crw_outputs/batch_files/crw_batch_052.csv
Starting batch 53 of 62
 Skipping already saved track: tuba tigre DJI_0442.SRT
 Skipping already saved track: tuba tigre DJI_0443.SRT
 Fitting track 3 in batch 53 : tuba tigre DJI_0444.SRT
 Skipping already saved track: tuba tigre DJI_0460.SRT
 Skipping already saved track: tuba tigre DJI_0461.SRT

Skipping already saved track: tuba tigre DJI_0462.SRT
Skipping already saved track: tuba tigre DJI_0463.SRT
Skipping already saved track: tuba tigre DJI_0464.SRT
Skipping already saved track: tuba tigre DJI_0465.SRT
Skipping already saved track: tuba tigre DJI_0466.SRT
Finished batch 53 - saved to: crw_outputs/batch_files/crw_batch_053.csv
Starting batch 54 of 62
Skipping already saved track: tuba tigre DJI_0519.SRT
Skipping already saved track: tuba tigre DJI_0521.SRT
Skipping already saved track: tuba tigre DJI_0522.SRT
Skipping already saved track: tuba tigre DJI_0523.SRT
Skipping already saved track: tuba tigre DJI_0525.SRT
Skipping already saved track: tuba tigre DJI_0526.SRT
Skipping already saved track: tuba tigre DJI_0527.SRT
Skipping already saved track: tuba tigre DJI_0584.SRT
Skipping already saved track: tuba tigre DJI_0586.SRT
Skipping already saved track: tuba tigre DJI_0587.SRT
Finished batch 54 - saved to: crw_outputs/batch_files/crw_batch_054.csv
Starting batch 55 of 62
Skipping already saved track: tuba tigre DJI_0629.SRT
Skipping already saved track: tuba tigre DJI_0630.SRT
Skipping already saved track: tuba tigre DJI_0737.SRT
Skipping already saved track: tuba tigre DJI_0738.SRT
Skipping already saved track: tuba tigre DJI_0739.SRT
Skipping already saved track: tuba tigre DJI_0740.SRT
Skipping already saved track: tuba tigre DJI_0741.SRT
Skipping already saved track: tuba tigre DJI_0742.SRT
Skipping already saved track: tuba tigre DJI_0745.SRT
Skipping already saved track: tuba tigre DJI_0746.SRT
Finished batch 55 - saved to: crw_outputs/batch_files/crw_batch_055.csv
Starting batch 56 of 62
Skipping already saved track: tuba tigre DJI_0747.SRT
Skipping already saved track: tuba tigre DJI_0748.SRT
Skipping already saved track: tuba tigre DJI_0749.SRT
Skipping already saved track: tuba tigre DJI_0750.SRT
Skipping already saved track: tuba tigre DJI_0752.SRT
Skipping already saved track: tuba tigre DJI_0753.SRT
Skipping already saved track: tuba tigre DJI_0754.SRT
Skipping already saved track: tuba tigre DJI_0755.SRT
Fitting track 9 in batch 56 : tuba tigre DJI_0756.SRT
Skipping already saved track: tuba tigre DJI_0758.SRT
Finished batch 56 - saved to: crw_outputs/batch_files/crw_batch_056.csv
Starting batch 57 of 62

Skipping already saved track: tuba tigre DJI_0772.SRT
Skipping already saved track: tuba tigre DJI_0773.SRT
Skipping already saved track: tuba tigre DJI_0774.SRT
Skipping already saved track: tuba tigre DJI_0776.SRT
Skipping already saved track: tuba tigre DJI_0777.SRT
Skipping already saved track: tuba tigre_DJI_0002_FB.SRT
Skipping already saved track: tuba tigre_DJI_0003_FB.SRT
Skipping already saved track: tuba tigre_DJI_0004_FB.SRT
Skipping already saved track: tuba tigre_DJI_0005_FB.SRT
Skipping already saved track: tuba tigre_DJI_0006_FB.SRT
Finished batch 57 - saved to: crw_outputs/batch_files/crw_batch_057.csv
Starting batch 58 of 62
Skipping already saved track: tuba tigre_DJI_0019_FB.SRT
Skipping already saved track: tuba tigre_DJI_0020_FB.SRT
Skipping already saved track: tuba tigre_DJI_0022_FB.SRT
Skipping already saved track: tuba tigre_DJI_0023_FB.SRT
Skipping already saved track: tuba tigre_DJI_0024_FB.SRT
Skipping already saved track: tuba tigre_DJI_0025_FB.SRT
Skipping already saved track: tuba tigre_DJI_0026_FB.SRT
Skipping already saved track: tuba tigre_DJI_0027_FB.SRT
Skipping already saved track: tuba tigre_DJI_0479.SRT
Skipping already saved track: tuba tigre_DJI_0480.SRT
Finished batch 58 - saved to: crw_outputs/batch_files/crw_batch_058.csv
Starting batch 59 of 62
Skipping already saved track: tuba tigre_DJI_0481.SRT
Skipping already saved track: tuba tigre_DJI_0483.SRT
Skipping already saved track: tuba tigre_DJI_0485.SRT
Skipping already saved track: tuba tigre_DJI_0486.SRT
Skipping already saved track: tuba tigre_DJI_0776_FB.SRT
Skipping already saved track: tuba tigre_DJI_0778_FB.SRT
Skipping already saved track: tuba tigre_DJI_0779_FB.SRT
Skipping already saved track: tuba tigre_DJI_0780_FB.SRT
Skipping already saved track: tuba tigre_DJI_0781_FB.SRT
Skipping already saved track: tuba tigre_DJI_0782_FB.SRT
Finished batch 59 - saved to: crw_outputs/batch_files/crw_batch_059.csv
Starting batch 60 of 62
Skipping already saved track: tuba tigre_DJI_0783_FB.SRT
Skipping already saved track: tuba tigre_DJI_0784_FB.SRT
Skipping already saved track: tuba tigre_DJI_0976_FB.SRT
Skipping already saved track: tuba tigre_DJI_0977_FB.SRT
Skipping already saved track: tuba tigre_DJI_0978_FB.SRT
Skipping already saved track: tuba tigre_DJI_0979_FB.SRT
Skipping already saved track: tuba tigre_DJI_0980_FB.SRT


```

Skipping already saved track: tuba tigre_DJI_0981_FB.SRT
Skipping already saved track: tuba tigre_DJI_20241029153807_0025_D_FB.SRT
Skipping already saved track: tuba tigre_DJI_20241029154040_0026_D_FB.SRT
Finished batch 60 - saved to: crw_outputs/batch_files/crw_batch_060.csv
Starting batch 61 of 62
  Skipping already saved track: tuba tigre_DJI_20241029154253_0027_D_FB.SRT
  Skipping already saved track: tuba tigre_DJI_20241029154514_0028_D_FB.SRT
  Skipping already saved track: tuba tigre_DJI_20241029154558_0029_D_FB.SRT
  Skipping already saved track: tuba tigre_DJI_20241029154638_0030_D_FB.SRT
  Skipping already saved track: tuba-tigre_DJI_0924_FB.SRT
  Skipping already saved track: tuba-tigre_DJI_0925_FB.SRT
  Skipping already saved track: tuba-tigre_DJI_0927_FB.SRT
  Skipping already saved track: tuba-tigre_DJI_0928_FB.SRT
  Skipping already saved track: tuba-tigre_DJI_0929_FB.SRT
  Skipping already saved track: tuba-tigre_DJI_0930_FB.SRT
Finished batch 61 - saved to: crw_outputs/batch_files/crw_batch_061.csv
Starting batch 62 of 62
  Skipping already saved track: tuba-tigre_DJI_0931_FB.SRT
  Skipping already saved track: tuba-tigre_DJI_0932_FB.SRT
Finished batch 62 - saved to: crw_outputs/batch_files/crw_batch_062.csv

```

```

batch_files <- list.files(
  "crw_outputs/batch_files",
  pattern = "^crw_batch_.*\\.csv$",
  full.names = TRUE
)

gps_crw_all <- batch_files %>%
  map_dfr(read_csv, show_col_types = FALSE)

```

```

meta_tide_moon <- read_csv(file.path(project_dir,"srt_metadata_with_tide_moon.csv"))

gps_crw_all <- gps_crw_all %>%
  dplyr::mutate(file_name = as.character(file_name)) %>%
  dplyr::select(-dplyr::any_of(c(
    "datetime_srt", "datetime_local",
    "year", "year.x", "year.y",
    "month_num", "month_num.x", "month_num.y",
    "month", "month.x", "month.y",
    "file_date", "file_date.x", "file_date.y"
  ))) %>%
  dplyr::left_join(meta_tide_moon, by = "file_name") %>%

```

```

dplyr::mutate(
  year = lubridate::year(datetime_local),
  month_num = lubridate::month(datetime_local),
  month = factor(
    lubridate::month(datetime_local, label = TRUE, abbr = TRUE),
    levels = month.abb,
    ordered = TRUE
  ),
  correction_dist_m = sqrt((x_true_m - x_obs)^2 + (y_true_m - y_obs)^2)
)

# Save
write_csv(gps_crw_all, "crw_outputs/gps_2s_clean_crw_all_tracks_enriched.csv")

# =====
# LOAD CLEAN DATASET
# =====
gps_crw_all <- readr::read_csv(
  "crw_outputs/gps_crw_all_enriched_FINAL.csv",
  show_col_types = FALSE
)

# =====
#Track-level summary
track_summary <- gps_crw_all %>%
  group_by(file_name) %>%
  summarise(
    n = n(),
    year = first(year),
    month = first(month),
    tide_height_m = first(tide_height_m),
    tide_state = first(tide_state),
    moon_phase = first(moon_phase),
    moon_illumination = first(moon_illumination),
    mean_correction = mean(correction_dist_m, na.rm = TRUE),
    median_correction = median(correction_dist_m, na.rm = TRUE),
    max_correction = max(correction_dist_m, na.rm = TRUE),
    .groups = "drop"
  )

write_csv(track_summary, "crw_outputs/crw_track_summary.csv")

```

```
gps_crw_all %>%
  summarise(
    mean_correction = mean(correction_dist_m, na.rm = TRUE),
    median_correction = median(correction_dist_m, na.rm = TRUE),
    max_correction = max(correction_dist_m, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 3
  mean_correction median_correction max_correction
      <dbl>          <dbl>          <dbl>
1    9589003.        9589015.        9589274.
```

```
write_csv(gps_crw_all, "crw_outputs/gps_crw_all_enriched_FINAL.csv")
```

```
##data integrity check
```

```
gps_crw_all %>%
  dplyr::count(year, month) %>%
  tidyr::pivot_wider(names_from = month, values_from = n, values_fill = 0)
```

```
# A tibble: 5 x 12
  year   Nov   Dec   Feb   Jan   Mar   May   Aug   Jul   Jun   Oct   Sep
  <dbl> <int> <int> <int> <int> <int> <int> <int> <int> <int> <int> <int>
1  2022  1749     0     0     0     0     0     0     0     0     0     0
2  2023  3337   477  2287  1410   123   624     0     0     0     0     0
3  2024     0     0   365  3617     0     0  1273   811   670   494  1029
4  2025  4320  1024     0     0     0     0     0     0     0     0     0
5  2026     0     0  1541  1256  1010     0     0     0     0     0     0
```

Combine and summarise

```
nrow(gps_crw_all)
```

```
[1] 27417
```

```
n_distinct(gps_crw_all$file_name)
```

```
[1] 592
```

```
head(track_summary)
```

```
# A tibble: 6 x 11
  file_name          n year month tide_height_m tide_state moon_phase
  <chr>          <int> <dbl> <chr>          <dbl> <chr>      <chr>
1 20231226_tuba tigre_DJI~    99  2023 Dec           2.11 rising    Full
2 20231226_tuba tigre_DJI~    99  2023 Dec           2.16 rising    Full
3 20231226_tuba tigre_DJI~    99  2023 Dec           2.16 rising    Full
4 20231226_tuba tigre_DJI~    99  2023 Dec           2.16 rising    Full
5 20231226_tuba tigre_DJI~    74  2023 Dec           2.21 rising    Full
6 20231226_tuba tigre_DJI~     7  2023 Dec           2.21 rising    Full
# i 4 more variables: moon_illumination <dbl>, mean_correction <dbl>,
#   median_correction <dbl>, max_correction <dbl>
```

```
summary(track_summary$phi)
```

```
Length Class Mode
      0  NULL  NULL
```

```
summary(track_summary$sigma_gps)
```

```
Length Class Mode
      0  NULL  NULL
```

```
summary(track_summary$sigma_move)
```

```
Length Class Mode
      0  NULL  NULL
```

```
summary(track_summary$mean_correction)
```

```
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
9588577 9588854 9588965 9588967 9589085 9589273
```

```
summary(track_summary$median_correction)
```

```
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
9588577 9588854 9588965 9588967 9589085 9589273
```

```
summary(track_summary$max_correction)
```

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	9588577	9588854	9588965	9588968	9589085	9589274

Kernel density estimation in projected space (meters)

```
library(dplyr)
library(sf)
library(terra)
library(MASS)
library(ggplot2)
library(tidyterra)

turbo_dark <- c(
  "#30123B",
  "#2C7FB8",
  "#4AC16D",
  "#F2C14E",
  "#C81D11"
)

# make sure projected corrected coordinates exist
# use gps_crw_all for corrected positions
x_range <- range(gps_crw_all$x_true_m, na.rm = TRUE)
y_range <- range(gps_crw_all$y_true_m, na.rm = TRUE)

kde_m <- MASS::kde2d(
  x = gps_crw_all$x_true_m,
  y = gps_crw_all$y_true_m,
  n = 300,
  h = c(25, 25),
  lims = c(x_range, y_range)
)

kde_m_df <- expand.grid(
  x = kde_m$x,
  y = kde_m$y
)
kde_m_df$z <- as.vector(kde_m$z)
```

```

vals_m <- kde_m_df$z
q_m <- quantile(vals_m, probs = seq(0, 1, length.out = 6), na.rm = TRUE)

q_m[1] <- min(vals_m, na.rm = TRUE) - 1e-12
q_m[length(q_m)] <- max(vals_m, na.rm = TRUE) + 1e-12

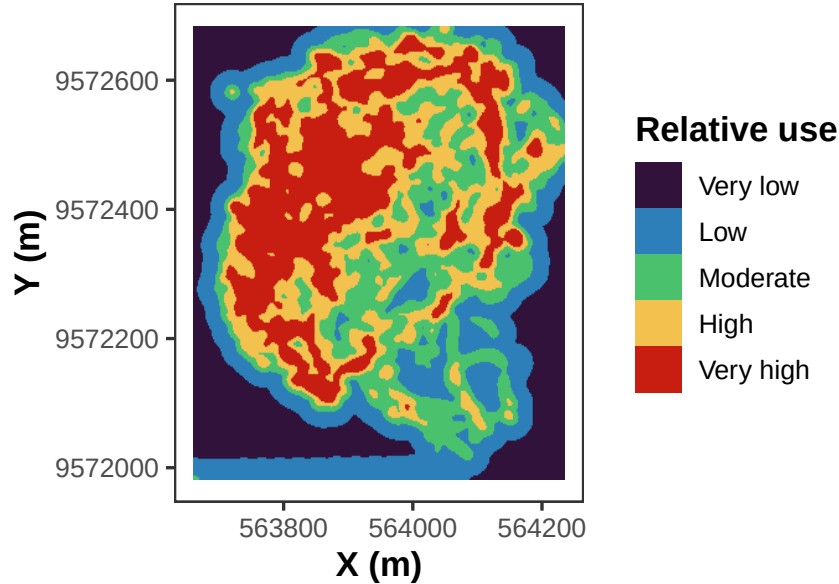
kde_m_df <- kde_m_df %>%
  mutate(
    use_class = cut(
      z,
      breaks = q_m,
      include.lowest = TRUE,
      labels = c("Very low", "Low", "Moderate", "High", "Very high")
    )
  )

p_hotspot_overall <- ggplot(kde_m_df, aes(x = x, y = y, fill = use_class)) +
  geom_raster() +
  coord_equal() +
  scale_fill_manual(
    values = turbo_dark,
    drop = FALSE,
    name = "Relative use"
  ) +
  theme_bw(base_size = 13) +
  theme(
    panel.grid = element_blank(),
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(face = "bold"),
    legend.title = element_text(face = "bold"),
    legend.text = element_text(size = 10)
  ) +
  labs(
    x = "X (m)",
    y = "Y (m)",
    title = "Tiger shark core-use areas"
  )

p_hotspot_overall

```

Tiger shark core-use areas



Kernel density estimation overlayed on basemap hotspot figure

```
turbo_dark <- c(
  "#30123B",
  "#2C7FB8",
  "#4AC16D",
  "#F2C14E",
  "#C81D11"
)

# -----
# 1. KDE in projected space (meters)
# -----

x_range_m <- range(gps_crw_all$x_true_m, na.rm = TRUE)
y_range_m <- range(gps_crw_all$y_true_m, na.rm = TRUE)

kde_m <- MASS::kde2d(
  x = gps_crw_all$x_true_m,
  y = gps_crw_all$y_true_m,
  n = 300,
  h = c(25, 25),
```

```

    lims = c(x_range_m, y_range_m)
  )

# -----
# 2. Convert KDE grid to raster in UTM 25S
# -----
kde_m_df <- expand.grid(
  x = kde_m$x,
  y = kde_m$y
)
kde_m_df$z <- as.vector(kde_m$z)

kde_rast_utm <- rast(
  kde_m_df,
  type = "xyz",
  crs = "EPSG:32725"
)

names(kde_rast_utm) <- "density"

# -----
# 3. Reproject to lon/lat
# -----
kde_rast_ll <- terra::project(kde_rast_utm, "EPSG:4326")
kde_rast_ll[kde_rast_ll < quantile(values(kde_rast_ll), 0.05, na.rm = TRUE)] <- NA

# -----
# 4. Convert to data frame for plotting
# -----
# trim only the weakest background
vals <- terra::values(kde_rast_ll)[, 1]
thresh <- quantile(vals[vals > 0], 0.15, na.rm = TRUE)

kde_trim <- kde_rast_ll
kde_trim[kde_trim < thresh] <- NA

# convert to data frame and drop NA cells
kde_df_plot <- as.data.frame(kde_trim, xy = TRUE, na.rm = TRUE)
names(kde_df_plot)[3] <- "density"

##Make SF object
gps_2s_sf <- gps_2s_clean %>%

```



```

    st_as_sf(coords = c("longitude", "latitude"), crs = 4326, remove = FALSE)

# Add satellite basemap
bbox_sf <- st_as_sf(st_bbox(gps_2s_sf)) %>%
  st_buffer(0.0008)

sat_map <- get_tiles(
  x = bbox_sf,
  provider = "Esri.WorldImagery",
  crop = TRUE,
  zoom = 17
)

## KDE basemap
p_basemap_context <- ggplot() +
  geom_spatraster_rgb(data = sat_map) +
  geom_raster(
    data = kde_df_plot,
    aes(x = x, y = y, fill = density),
    alpha = 0.65
  ) +
  coord_sf(
    xlim = st_bbox(bbox_sf)[c("xmin", "xmax")],
    ylim = st_bbox(bbox_sf)[c("ymin", "ymax")],
    expand = FALSE
  ) +
  scale_fill_gradientn(
    colours = turbo_dark,
    values = scales::rescale(c(0, 0.05, 0.15, 0.40, 1)),
    name = "Relative space use",
    breaks = c(
      min(kde_df_plot$density, na.rm = TRUE),
      max(kde_df_plot$density, na.rm = TRUE)
    ),
    labels = c("Low", "High"),
    guide = guide_colorbar(
      reverse = TRUE,
      barheight = unit(3.5, "cm"),
      barwidth = unit(0.4, "cm")
    )
  ) +
  theme_void(base_size = 13) +

```

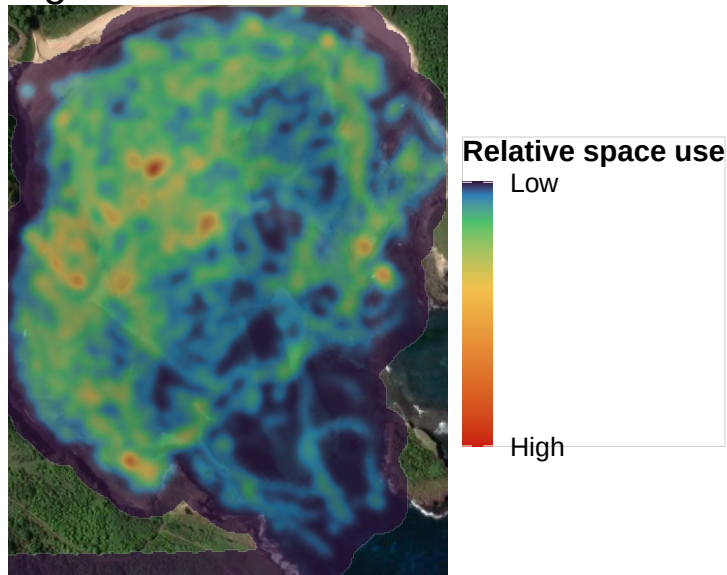
```

theme(
  legend.position = "right",
  legend.key.height = unit(0.5, "cm"),
  legend.key.width = unit(0.4, "cm"),
  legend.title = element_text(size = 10.5, face = "bold"),
  legend.text = element_text(size = 9.5),
  legend.background = element_rect(fill = "white", colour = "grey80")
) +
labs(
  title = "Tiger shark core-use areas",
  caption = "Basemap: Esri World Imagery"
)

p_basemap_context

```

Tiger shark core-use areas



Basemap: Esri World Imagery

““